

10 The saxophone today

The contemporary saxophone

CLAUDE DELANGLE AND JEAN-DENIS MICHAT (TRANSLATED BY PETER NICHOLS)

In praise of saxophonists

With so-called traditional instruments, the gradual emergence of major works and an evolution of repertoire have, over time, contributed to the steady development of a particular school, usually with consistent and well-defined concepts of writing, phrasing, tone and stylistic parameters – in other words, a clear notion of what represents ‘good’ playing.

Saxophonists themselves have long faced the thorny problem of repertoire, both in terms of quality and quantity. Although orphaned for the sins of its youth, the saxophone is currently enjoying one of its most fruitful phases, with the emergence of a stability not dissimilar to that being experienced by contemporary composers.

The fifty years since the Second World War have seen both iconoclastic and purely exploratory adventures: the blind alley of total serialism and the experimental electronic ideas of the 1980s. Today’s young composers see themselves as educated but not ‘academic’, knowledgeable but still inquisitive, displaying both brilliance and perception. The same applies to saxophonists. Free from any musical ‘establishment’, they have a ferocious appetite for new music and an acute sense of responsibility which will undoubtedly benefit the generations of the twenty-first century. Very rarely tied to one style, method or particular practice, their cultural base, while not actually larger than that of other instrumentalists, is often much more receptive to the most stimulating of influences. Coming from a state of ‘No Fixed Repertory’ the saxophonist can often end up trying everything, as much a polymath as his instrument is a polymorph. He plays all the instruments of the family, teaches and fits easily into the most diverse ‘socio-musical classes’. The development of a student reflects this: basic technique (Boehm system) is based on the concept of ‘everything is playable’. As nineteenth-century saxophone music is virtually nonexistent, the student has

hardly known the type of repertoire in which the fingers trust the ear. Instead, he is brought up from the start in an atonal, often extremely chromatic, language. Melodically tortuous lines sustained by unstable harmonies in fast tempi can over-exaggerate the importance of technical dexterity. Once the study repertoire of the flute, oboe and clarinet (Marcel Moyse, Marcel Mule) has been absorbed (usually rapidly) the next step, apart from the specific studies for the instrument, is to tackle violin music (with Paganini *Caprices* at the top of the list!).

All improvised music (jazz, ethnic etc.) is an integral part of the modern saxophonist's culture. Even if this is not played regularly, the cord which binds the classical performer to the notes has been irrevocably severed, along with the phrasing and all the formal 'limitations' which can characterise the concert in its inherited nineteenth-century form.

The nature of the saxophone family has given rise to a large number of transcriptions, so creating as broad a repertoire as possible: oboe, violin, flute on the soprano saxophone; clarinet, viola, French horn on the alto saxophone; viola da gamba, French horn on the tenor saxophone; bassoon, cello on the baritone saxophone.

With all this wide experience, the saxophonist can sympathise with the dilemma of the contemporary composer – how to build on quasi-virgin territory? All saxophonists today are interested in the creative process, although obviously each one has their own stylistic preferences. New works are often the result of artist–composer relationships and can reflect the personality of the dedicatee.

In America, John Samplen has been responsible for works by Shrode, Beerman, Francesconi, Mabry, Subotnick, Martino, Glass, Harbison, Furman, Babbitt, Cage, Tower, Noda, Stockhausen and Mays. Kenneth Radnofsky collaborates with many of the above as well as with Bell, Theofanidis, Schüller, Sims and Korde; he is also the inspiration behind the World Wide Concurrent Premières and Commissioning Fund, Inc., a world association of saxophonists promoting new music; William H. Street, Paul Bro, Kenneth Fischer, the Chicago Saxophone Quartet and many others are constantly generating new repertoire.

In France, Jean-Marie Londeix has worked extensively with composers, notably a young generation taught by Fusté Lambezat in Bordeaux (Lauba, Rossé and Havel). His successor Marie Bernadette Charrier has also inspired many works (Havel, Lauba, Melle and Rossé). Daniel Kientzy has turned

towards electro-acoustic music in collaboration with the Groupe de Recherche Musicale; and Teruggi with Romanian composers Anatole Vieru, Stefan Niculescu, Myriam Marbé and Costin Mioreanu. A number of saxophone quartets commission composers: Xasax (Barry Guy, Alex Buess, Alvaro Carlevaro, Bernardo Kuczer and Denis Levaillant); the Diastema Quartet work with Dubedout, Martin, Level and Burgan. Serge Bertocchi, outside of his work with Xasax and Ars Gallica, is developing an important repertoire for the baritone saxophone: De Clerck, Bruno Giner (*Yod*), Henry Kergomar (*Vianässa*) and Marie Helene Fournier (*Hyppogriffe*).

In Italy, Federico Mondelci, a composer and performer, Alberto Domizi and the Academia Saxophone Quartet promote their country's composers (Scogna, Rossi Re, Tesei and Sbordonì). In Germany, Johannes Ernst works with well-known composers (Kyburz, Staude, Mundry and Zimmermann). The American saxophonist John Edward Kelly and the Rascher Quartet are also responsible for a lot of creative activity in Germany; *Rasch* by Franco Donatoni, written for them, is heavily influenced by Latin poetry and displays astonishing rhythmic vitality in its classical writing. In Russia, Alexander Volkov; in the Netherlands, Arno Bornkamp and Henry Bok; in Belgium, Hans de Jong (also a composer); in England, John Harle (with composers Muldowney and Birtwistle); and in Cuba, the saxophonist Villafruela. In Scandinavia, the New Danish Saxophone Quartet are building a substantial Danish repertoire (Lorentzen, Norholm, Hentze and Roikjer).

In Croatia the Zagreb Quartet perform music by Papandopulo, Ruzdjak, Bjelinski and Detoni, combining the Croatian folk tradition with twelve-tone or post-classical principles. Manuel Mijan and Francisco Martinez Garcia work closely with the Spanish composers (Tomas Marco, Rojo, Angulo, Alis, Seco and Louis De Pablo). In Finland, the saxophonist Pekka Savijoki has commissioned a number of talented composers, among them Magnus Lindberg and Esa-Pekka Salonen, to produce concertos, solo works and chamber music. In Austria, the Vienna Quartet are putting together an interesting and high quality repertoire (Engebretson, Urbanner and Wagner). In Switzerland, Marcus Weiss, the successor to Iwan Roth, has inspired the creation of a substantial solo repertoire (Sotelo, Haubenstock-Ramati and Walter Zimmermann) and works for the Acantus Trio (saxophones, piano and percussion). In Japan, the Quatre Roseaux Quartet and soloist Nobuya Sugawa have premièred many new works; the saxophonist composer Ryo

Noda has developed an important repertoire for his instrument, principally for solo saxophone.

The vocal saxophone

Invented by Adolphe Sax around 1840, the saxophone was intended to be an outdoor instrument combining the phrasing and tone of stringed instruments with the power of brass instruments. The result was undoubtedly a success, except, far from being an instrument of compromise, it actually accumulated the respective qualities: the powers of expression, attack and graininess of the strings come naturally to it, as do the breadth, volume and roundness of the brass. Its dynamic palette is enlarged further by the intrinsic qualities of a single-reed instrument which endow it with the extremely soft dynamics of its clarinet cousin. It lacks only words as the sound is, with its flexibility and expressiveness, the closest to the human voice (not for nothing is it the ‘voice of jazz’). With a concentration of harmonics at around 2000 Hz (the optimal zone of acute hearing) it is very similar to the human voice.

It is significant that the saxophone was introduced into vocal music in such a natural way. One of the first notable examples is the use of the alto saxophone to unite, sustain and guide the children’s choir in Puccini’s *Turandot*. The timbre obtained is highly effective, the flexibility of the instrument being so close to the purity of the children’s voices that the ensemble produces an intriguing and hitherto unprecedented texture. It is also interesting to observe that a great number of composers attracted to the human voice have instinctively adopted the instrument, even using it exclusively for characters in their dramatic works. Indeed, the credit for introducing the saxophone into the orchestra should go to opera, and not to any symphonic tradition.

The Italians lead the way in employing the saxophone to good effect: Berio with *Air Aus* (tenor), *Traces* (three saxophones), *Un Re In Ascolto* (alto), *La Vera Storia* (two altos) and *Outis*, performed in Milan in 1996 with full quartet (SATB). When doubling the choir strictly it thickens the texture, and when underscoring it helps the voice and also gives discreet intonation support. Dallapiccola’s music is a natural home for the instrument – *Il prigionero* (tenor, alto), *Ulysse* (two altos), *Canti di liberazione* (two saxophones) – as is Maderna’s: *Don Perlimplin* (five saxophones). If it seems to be a second skin in vocal music, it is an equally indispensable tool

of orchestration with its capacity to integrate with, and forge an active link between, all instrumental groupings. Consider its place in the dramatic works of the Second Viennese School: Berg's *Lulu* (alto), Schoenberg's *Von Heute auf Morgen* (soprano/alto, tenor/bass) and Webern's *Das Augenlicht*, Op. 26. Its strong tonal identity is required here, an identity that can even make social statements as in Stockhausen's *Licht*, Zimmermann's *Die Soldaten* or Henze's *Das Floss der 'Medusa'*.

In intimate settings the saxophone blends supremely, its considerable dynamic range (from *pppp* to *fff* in Berio), its ease of glissando (to the sixth harmonic in the top register), and its palette of colours and timbres make it a popular chamber music partner. Take for example *Plupart du temps II* by Jolas for tenor saxophone, tenor voice and cello; Raskatov's *Pas de deux* for tenor and soprano saxophones and soprano voice; Scelsi's *Yamaon* for bass voice, alto and baritone saxophones, contrabassoon, double-bass and percussion; and Berio's *Prière* for voice and eight instruments. Perhaps the saxophone is even beginning to supplant the clarinet in its role as the instrumental double of the voice. When not actively replacing it, it can function as a textural and dynamic extension: *Schlimann* by Jolas (clarinet and tenor sax), Berio's *Canticum Novissimi Testamenti* for eight voices, four saxophones and four clarinets. Any analysis of the saxophone–voice relationship would be incomplete without mentioning the dramatic effect of the instrument in Pousseur's *Votre Faust*. In France, Georges Aperghis, a composer of Greek origin, has written *Simplexis* (two altos, two tenors, baritone) and *Signaux* for four instruments of the same range. The latter, a strangely hypnotic work, is based on three tirelessly reproduced series of eight quarter-tones and has parallels with the work of John Cage. The composer Alain Louvier in *Le Jeu des sept musiques* sets several styles against each other with a particularly attractive scenario in which rock music triumphs! Similar in philosophy to John Cage, the Dutchman Eric De Clerck displays an interesting use of space in his solo baritone piece *Why?* In Kagel's *Zwei Akte* for alto saxophone and harp the instrumentalists do not actually move but dramatic action takes place within the music. This natural extroversion, the principal reason behind suspicion of the instrument from some quarters, is but one facet of the saxophone's personality, its polymorphic nature having led it naturally towards solo performance.

Certain works require the simultaneous playing of two saxophones. If the use of four or more saxophones in the course of a piece achieved a certain

success in the 1980s (Jolas, Marbé, De Pablo and Donatoni), composers today seem to have gone back to using one or two. This latter solution employed by Raskatov (*Pas de Deux*: soprano and tenor) and Jolas (*Lumor*: soprano and tenor) seems to be an interesting middle course allowing the development of the soloist without turning him/her into a furniture-mover or musical freak-show!

The origins of contemporary technique

When Adolphe Sax was displaying his saxophones, creating a repertoire, playing or teaching, his priority was to demonstrate the revolutionary acoustical qualities of his invention. Beautiful lyrical melodies and ‘showy’ variations from the early works illustrate the essential qualities of all the family. The repertoire built by Elise Hall at the turn of the century reveals the expressive potential of the instrument in much greater depth through its phrasing, projection and dynamic variety.

The classical schools subsequently began to develop a high level of virtuosity; in Berlin, Bumcke analysed the fundamentals of technique and introduced the concept of altissimo (*Saxophon Schule*, Edition Benjamin, 1926), and Schoenberg used bass and tenor saxophones with altissimo as early as 1930 in the orchestra of *Von Heute auf Morgen*. Thereafter, the altissimo was widely developed by Sigurd Rascher, based on the natural harmonics of the low register.

Under the influence of Rascher, composers such as Frank Martin, Jacques Ibert, Karel Husa and Henry Cowell made use of a four-octave saxophone. The American school carried on this tradition: Ted Nash (*High Harmonics*, 1946), Rosemary Lang (*Altissimo Register*) and Eugene Rousseau (*Saxophone High Tones*). In this respect, France did not make up its lost ground until the 1970s: Denisov (*Sonata 1970* written for Londeix) and Constant (*Concertante*, 1978). Until then Marcel Mule, the founder of the French school, had little interest in this aspect. Instead, through his teaching, his didactic works and his transcriptions (Mule–Leduc) he contributed greatly to widespread improvements in sound, dynamics, tonal homogeneity, virtuosity of articulation and velocity. Originally a violinist (as was his successor at the Paris Conservatoire, Daniel Deffayet), he began with an analysis of the violin and brought to the saxophone aspects of string style. The origins of saxophone vibrato can also be traced back to this.

Through their analyses of acoustic phenomena, approaches to technical improvement and their graded studies, early performers and teachers paved the way for an explosion of techniques seen in today's repertoire. Working with composers creates a repertoire ever closer to the true acoustical nature of the instrument, although the reverse can be true – difficulties encountered in a work can inspire an objective analysis of the problems and lead to the creation of appropriate exercises. Certain artists are notable for their work in codifying techniques: Larry Teal (in *The Art of Saxophone Playing*) tackles all the main points of technique, including the more advanced aspects (such as altissimo and double-tonguing). Donald Sinta has produced a video in which techniques are demonstrated with great virtuosity. In Japan, Sakaguchi has laid the foundations for solid technical training, producing players of the calibre of Omuro, Sugawa and Muto. In the USA, Eugene Rousseau and Frederick Hemke have contributed to the development of an American repertoire by composers such as Muczynski, Creston, Stein, Heiden and Karlins. In France, Jean-Marie Londeix has written numerous works: *Hello! Mr Sax* and *Paramètres du Saxophone* (Leduc) deal with different parameters of sound (pitch, timbre, duration, volume and attack), with corresponding charts.



Fig. 10.1 Nobuya Sugawa, Fourth British Saxophone Congress, 1996

Star quality

The instrument has been a success in jazz where, as previously mentioned, the hybrid nature of its tone is underlined. For jazz musicians the saxophone is a member of the brass family, whereas for classical players it is a woodwind. A saxophonist is recognised to a greater extent than any other instrumentalist by personal sound quality. For so-called 'academic' composers, the history of jazz and its saxophonists has long been a unique reference point when writing for the instrument. It is clear that the baggage of the modern saxophonist is the legacy and accumulation, as much as the synthesis, of all the modes of playing, of sounds and of styles developed by Charlie Parker, John Coltrane, Ornette Coleman, Dave Liebman etc. Of all the jazz techniques, only subtone (Ben Webster, Lester Young, Coleman Hawkins) seems to have originated in jazz itself. Jazz saxophone has virtually developed in parallel with the contemporary saxophone: the young generation make great use of multiphonics (Michael Brecker), slap-tonguing, circular breathing (James Carter) and double- and triple-tonguing (Louis Sclavis).

These parallels are sometimes openly acknowledged as composers attempt to capture on paper an improvised style. For example, *Opcit* by Hurel for tenor saxophone, where the manipulation of sound is a key factor (harmonics, growling etc.); *Hard* by Lauba for tenor saxophone which uses multiphonic rhythms as an analogy to the layered sounds of heavy rock; Denisov's *Sonata* for alto and piano, a 'classic' of the repertoire with a finale influenced very firmly by post-bop and perfectly written for the instrument. The fact that a Ben Webster, a Sonny Rollins or a Michael Brecker can draw from the same 'tool' such disparate sounds demonstrates the extraordinary tonal resources of the instrument, and above all its flexibility. This propensity for metamorphosis has given rise to a number of works with tape, for example Cavanna's *Goutte d'or blues* (soprano, sopranino and tape) and Zbar's *Jazzy Night in Yellow* (tenor and tape).

Some jazz saxophonists have written for the instrument themselves: Phil Woods gives a personal touch to be-bop with repetitive figures and irregular accents (*Sonata, Three Improvisations for Quartet*); François Jeanneau writes in a way which preserves the spontaneity of improvisation (*Suite for Quartet, Une Anche Passe*); David Liebmann fits together complex rhythms and silences in an unpredictable way (*A Moody Time*); tonal freedom, thematic fragmentation and textural juxtapositions are characteristic of

Martial Solal (*Une Pièce pour Quatre*); the sense of pulse is a determining factor in the quartet *Attractions* by Denis Levaillant.

In his concerto *For Stan Getz*, Richard Rodney Bennett combines jazz harmonies with free twelve-tone technique: he mixes the genres without a trace of compromise. The blues feature in concertos by Stanley Myers and the Flemish composers René Pieper and Kees Schoonebeek; whilst Michael Torke writes short, memorable themes, not unlike advertising jingles, in an attempt to make his music accessible. Martijn Padding mixes be-bop and classical styles within a more abstract form (*Quatuor Rotorno*); Tristan Keuris (*Music for Saxophone*) is not afraid of tonal references even if he sometimes uses harsh and strident sounds as a reaction against the lyrical style of the classical saxophone quartet. Geert Van Keulen uses four saxophones as a big band, and the distorted sounds of free jazz – use of the voice, shouts, coarse flutter-tonguing, playing with the teeth on the reed – form the basis of the compositional language of the Swiss composer Alex Buess (*Hyper Baton*). Unusual jazz improvisation sequences and untempered notes are the hallmark of Paul Méfano in *Périple*s (1978), which takes the natural tone of the tenor as its starting point, whereas jazz-rock is the central theme of *Anu Anvil* and *Parachutes de Dror Feiler*.

Some other works reflect not so much the jazz identity of the saxophone but rather the ethnic connotations of its improvisatory character: *Aulodie* by Mâche (soprano and tape) is originally for oboe da caccia, but is more convincing on the soprano. The popular roots of the instrument and its appropriation of the most diverse traditions – Indian music, Balkan music, South American music etc. – have influenced many composers. Based in Berlin since 1982, the Filipino composer Conrado Del Rosario, a member of a gamelan ensemble, wrote *Ringkaran* in 1990, the title inspired by South Philippine religious chants. Christian Lauba synthesises Western music with North African music – the music of his childhood – and Japanese music. *Les Sept Isles* for saxophone ensemble is a true meeting of cultures out of which the composer produces a sophisticated mosaic; his violent and extrovert solo saxophone work *Hard* is influenced by rock music while *Steady Study on the Boogie* (1993) reveals its roots in the title. The American composer William Albright introduces elements of popular American music in the *Scherzo* of his post-classical Sonata (1984). A chaconne variation shows the influence of Baroque music.

Japanese composers refer to their traditional music in different ways: descriptions of nature are to be found by Takashi Yoshimatsu in *Fuzzy Bird Sonata* and *Cyber Bird Concerto*; references to Japanese instrumental techniques in Ryo Noda (*Mai, Improvisations I, II, and III, Gen Concerto*), Joji Yuasa (*Not I but the Wind*) and Natsuda (*West*); references to the theatrical concepts of time and space in Yoshihisa Taïra (*Pénombres VI*), Toshio Hosokawa (*Verticale Time Study II*) and Toru Takemitsu (*Distance*, originally written for oboe, with optional sho [a Japanese mouth-organ] accompaniment). Takemitsu's conception of time is very pertinent: 'Whereas the modern Western concept of time is by nature linear, that is to say its duration always keeps the same state, in Japan, time is seen as an entity which continually repeats itself'.¹

The Romanian composer Stefan Niculescu uses a Byzantine melody as the basis of his *Symphonie Concertante No. 3 'Cantos'*, as well as elements of Gregorian and non-European music; in his *Concerto*, Myriam Marbé makes use of popular songs. John Harbison's *San Antonio Sonata*, although of a classical structure, is based on popular Mexican music, funk and be-bop.

As it gradually frees itself from jazz and popular influences, and asserts its independence, the modern saxophone, far from disowning its roots, will increasingly exploit its unique duality: aggressive or tender, refined or vulgar. Ever since composers learned how to quantify and exploit the uncertainties of 'real-time' creation, the modern classical saxophone school has produced artists able to master even the most transcendent of works.

There is a great deal of repertoire for solo saxophone, and as with other instruments, the great names stand out. Berio's two *Sequenze IXb* (alto, originally for clarinet) and *VIIb* (soprano, originally for oboe): apart from the intrinsic interest of these two pieces, it is revealing to compare the originals with the new versions and to identify their respective characteristics – the uniqueness of the clarinet legato and the alto saxophone cantabile; the even tone of the soprano and the inimitable sound of the oboe. They are not 'saxophone versions', but two idiomatic and autonomous masterpieces. Similarly with Stockhausen's *In Freundshaft* (alto or soprano, originally for clarinet but transcribed for all instruments by the composer) where the compositional process is continued by gestural evolution (the work is played from memory). *Anubis-Nout* by Grisey demonstrates the tonal richness of the bass saxophone, its capacity to isolate harmonics (a ghostly effect) and also the remarkable nature of a family whose deeper instruments have a

flexibility and speed that almost equal those of its higher members. Because of the progressive transformations and the subtly varied sequences employed, it is a demonstration of development through tiny transitions rather than through contrasts.

This love of tonal metamorphosis is exemplified in the work of Giacinto Scelsi, who displays the harmonic power of saxophone timbre not through different modes of playing but by the examination of sound in an expanded temporal space in *Tre Pezzi* (soprano or tenor), *Maknogan* (baritone) and *Ixor* (soprano). The influence of Tibetan and Indian music is discernible, particularly in the modal nature of the works. All of Scelsi's attention is centred on the properties of sound, energy and life itself. Very often his music is built on the struggle between rest and movement, with repeated long notes set against rapidly moving neighbours.

The expressiveness and agility of the instrument, the spirit and charisma of its advocates, as well as its symbolic 'seul contre tous' image, have made the saxophone so attractive and yet so difficult for composers: Jolas' *Fourth Episode* (tenor), De Pablo's *Oculto* (baritone or alto), Takemitsu's *Distance* (soprano), Campana's *Acting In* (alto and tenor), Glass's *Gradus* (soprano), Wildberger's *Portrait* (alto), Méfano's *Tige* (alto), Ballif's *Solfegietto 8* (alto) and Fournier's *Horoscope* (alto). As concerto-type pieces become increasingly common, perhaps Adolphe Sax's brilliant invention will enter the orchestra via the front door, the soloist's entrance. It is true that its hybrid character, its capacity to drown out, fade away or blend, to disappear then return transformed or, indeed, in an unrecognisable form, allows the realisation of any number of compositional fantasies: a saxophonist never says no! All permutations are viable: full orchestra, for instance Denisov's *Concerto* (alto and orchestra), Constant's *Concertante* (alto and orchestra), Jolas' *Lumor* (tenor, soprano and orchestra) and De Pablo's *1 Couleur*; string orchestra, for instance Rossé's *Triangle pour un souffle* (alto and twelve strings); and various combinations, for example Jarrell's *Résurgences* (alto/soprano and eight instruments) and *Points d'Or* by Betsy Jolas for one saxophonist playing SATB, and fifteen instruments.

The saxophone and electronics

The harmonic-rich sound of the saxophone lends itself well to endless transformations. Furthermore, the range of the instrumental family allows

the retention of each register's principal properties and reduces the need for sampling which could distort the signal. A varied repertoire, which is still little known, expanded rapidly in the 1980s, in particular Decoust's *Olos* (tenor and electro-acoustic diffusion), Kurtag's *Interrogation* (tenor, bass, real-time Electronic IRCAM and tape) and Radulescu's *Astray* (saxophones, prepared piano and tape). Tape replaces the piano in a sonata combination very effectively, thereby sparing it the frequent unbalanced duels and unhappy marriages, despite the undoubted successes of the Denisov *Sonata* (alto and piano), Rossé's *La Main dans le souffle* (alto and piano), Hurel's *Bacasax* (alto and piano), Savouret's *A flanc de Bozat* (soprano/tenor and piano), Nodaira's *Arabesque 3*, Taira's *Pénombres 6* and the numerous works of Tisé, Gastinel, Gotkovsky and Robert.

Music for saxophone and tape

One particular scenario is common in compositions for instrument and tape: the instrument played live with its own projection on tape – for instance Isabel Mundry's *Composition 1992* – often with the tape part varying and breaking up the saxophone material, as in François Bernard Mâche's *Aulodie* and *Ensemble Invisible* by Aurelio Samori.

Milton Babbitt, one of the fathers of electronic music in the United States, has written a number of works for pure electronics or for instrument and tape. *Images* (1979) explores the various relationships possible between an instrument (here alto, soprano and sopranino) and tape. In *Voilements* and *Saxatile*, Jean Claude Risset expands the parameters of conventional listening – it is a shrewd person who can separate the saxophone from its treatment by the computer. A galactic world, atoms, particles and infinitely small domains await us in *Exultitudes*, composed in the studios of the Groupe de Recherches Musicales de l'INA (Radio France) by Gilles Racot. Other substantial works stand out: by Burton Beerman (*Concerto One* for saxophone and tape), Bernard Cavanna (*Goutte d'Or Blues*), Costin Miereanu (*Do- Mi- Si- La- Do- Ré-*, *Variants-Invariants*), Steve Reich (*Vermont Counterpoint*) and Michel Zbar (*Jazzy Night in Yellow*).

Real-time transformations

This is sometimes a question of using space or creating resonance. Joji Yuasa uses two microphones in *Not I but the Wind*, one to spatialise the instrument and the other linked to an echo chamber. Mention can also be made of *Métaksax* by Anatol Vieru for saxophone and envelope transformer.

Some composers use far more sophisticated computerised systems. Following a preordained program, the computer reacts to and transforms the instrumentalist's sounds in real time. Reference points can be directly recognised by the machine in the course of the performance or when the player himself controls the program via a pedal: for most of the time an assistant technician/musician is required to balance the sound, oversee the smooth running of the changes or indeed 'play' the changes live. Although fascinating, such systems are cumbersome and last-minute breakdowns happen without warning. In France, the Groupe de Recherches Musicales de l'INA has introduced a real-time audio-digital 'SYTER' system, a programmable and interactive sound processor. Wholly digital, this system combines the recording, processing and control elements necessary for such high-quality work. It has been responsible for several saxophone works, including *Xatys* by Daniel Teruggi.

Morton Subotnick, a leader and pioneer of new musical technology, maintains the duality between the traditional instrument and machine in his concerto *In Two Worlds* (optional version with orchestra), in which the saxophonist controls the electronic and computer-transformed sounds with a pedal. In *Music for Saxophone and Electronics* by Pablo Furman, a wide variety of sounds serves as the basis for the electronic part. Other substantial electronic works include *Olos* by Michel Decoust, *Solo* and *Spiral* by Karlheinz Stockhausen and *S* by Maurisio Bisati. Some composers have juxtaposed video, tape and musical text as in *The Stillness* by the American Reynold Wiedenaar. Sometimes the electronic part is played live on a synthesizer with acoustic instruments remaining untreated, as in *Linker Augentanz* by Stockhausen and *Moments Rituels* by Ton That Tiet.

Chamber music

The use of the saxophone in chamber music has long been a strict family affair. The 'classical' quartet (soprano, alto, tenor, baritone) has all the qualities of a perfect ensemble, but few great composers have found it a source of inspiration. If the tonal homogeneity of the quartet was an

attraction for classical composers, our contemporaries have often seen it as a handicap, a strong and inviolable identity. The best is yet to come, with the exceptions of *Xas* by Xenakis, *Vue sur les jardins interdits* by Henri Pousseur, *Four5* by John Cage, *Le Jeu des Sept Musiques* by Alain Louvier and the quartets of Dufourt, Dubedout and Nodaira.

The saxophone ensemble (often three sopranos, one doubling on soprano, three or four altos, three tenors, two baritones and a bass) is in effect a three-dimensional quartet. Here the tonal inertia is actually a plus point, the sheer numbers allowing great textural variety with the colours changing in the manner of an organ. The many permutations of instruments, dynamics and numbers allow for rewarding orchestrations. Several first-rate works are beginning to appear: Stockhausen's *Linker Augentanz* (ensemble, synthesizer and percussion), Tisné's *Messe pour notre Temps* (ensemble), Tiet's *Moments Rituels II* (ensemble), Eloy's *Quattrocento* (ensemble), Escaich's *Le chant des Ténèbres* (solo soprano and ensemble), Lauba's *Mutation-Couleurs IV* (ensemble) and Mantovani's *La fuite et son Ombre* (solo soprano and ensemble).

For contemporary saxophonists the route to total integration, acceptance and recognition without classical connotations depends increasingly on chamber music with other instruments. Due to its new-found stability, composers can easily imagine the instrument as a part of their music. It is no longer an experimental, exotic curiosity but is gradually being recognised for its intrinsic qualities. There are some excellent precedents: Webern's *Quartett*, Op. 22 (clarinet, tenor sax, violin and piano), Denisov's *Sonata* (alto and cello) and *Concerto Piccolo* (alto and six percussion), Jolas' *Plupart du Temps II* (tenor, tenor voice and cello), Leroux's *Phonie Douce* (alto, oboe and cello), Lindberg's *Linea d'Ombrea* (alto, flute, guitar and percussion), Tiet's *Moments Rituels I* (tenor, percussion and synthesizer) and Gilbert Amy's *Le Temps du Souffle* (violin, tenor and trombone). The mixture with 'les autres' – winds, strings or percussion – gives the 'sound painter' one extra primary colour on the palette. The twenty-first century will undoubtedly see the blossoming of a new 'traditional' instrument.

Current techniques: control of dynamics

Dynamic control is obviously not a new technique. Composers have always appreciated the saxophone's wide range of dynamic possibilities. As a result,

the mastery of extreme contrasts, subtle intermediate shades and the softest dynamics has become obligatory for today's performer.

Changes of timbre

The classical school has taken finger dexterity to a virtuosic degree (for example, in the music of Ibert and Desenclos) and developed an awareness of sound from established sources, principally the Romantic violin and bel canto (Tomasi, Glazunov). Today, without losing this precious heritage, saxophonists are expanding the notion of the 'beau son' considerably by developing a sort of 'tonal virtuosity' adapted to the language of each composer.

Vibrato

Classical repertoire is distinguished by a generalised use of vibrato, but in contemporary music the use of this tonal ornament is less systematic. Many composers require a particular sound: at the opening of *Sequenza IXb* Berio writes 'senza vibrato'. Alain Louvier (*Cinq Ephémères*), Ryo Noda (*Improvisations, Mai*) and Stockhausen (*In Freundshaft*) note precisely what must be vibrated, the speed or amplitude, and the nature of the undulation, pitch vibrato (obtained by jaw movement) or intensity vibrato (variation of the air-speed). This usage is similar to ornamental Baroque vibrato which was considered to be separate from sonority.

Flutter-tonguing

A classic contemporary technique, flutter-tonguing is obtained by rolling the tongue against the palate. With the tongue forward it is clear and smooth, particularly in *piano/pianissimo* passages; with the throat it is coarser, but can be used delicately for soft dynamics and extreme registers.

Microtonal trills

Microtonal trills (bisbigliando) create different colourings of the sound without unduly affecting intonation. They are generally soft and subtle and can be very fast, for instance in Yoshihisa Taïra's *Pénombres 6* and Ichiro

Nodaïra's *Arabesque 3*. Berio constructs his *Sequenza VIIb* (soprano version) around a B which sounds gently throughout the piece as if emanating from another instrument or a synthesizer. Variations of timbre, multiphonics, different articulations and subtle dynamics create an electro-acoustic effect. The second part of Grisey's *Anubis* for solo bass saxophone is punctuated by timbral melodies that echo the pitched melodies. The different fingerings of each pitch are precisely notated. Francis Cournet's *Thésaurus du Saxophoniste* (published by Billaudot) gives a comprehensive list of fingerings.

Subtone

A surprisingly soft sound can be obtained in the low register with the use of subtone; lower-jaw pressure is replaced by the tongue under the lip, or the tongue lightly touching the reed (as if pronouncing the word 'the'). Alternatively, the embouchure can be moved towards the end of the mouthpiece, holding it only with the lips and without pressure from the teeth. In each case the reed is partially prevented from vibrating and the upper partials are subdued: for use of this technique, see Paul Méfano's *Périple*, Jolas' *Fourth Episode* and Nodaïra's *Arabesque 3*.

Breath noise

As on the flute, the breath of the performer can mix with or replace the sound. In *Acting In* for solo tenor, the Argentine composer José Luis Campana notes precisely the quantity and proportion of the real sound and the breath: breath only/breath and sound/normal sound. François Rossé uses similar methods in his solo alto work *Le Frêne Egaré*. Sometimes the amount of breath and sound is left to the player. In *Not I but the Wind*, Joji Yuasa gives indications such as 'airy sound' or 'like a shakuhachi sound' or 'pure sound', invoking the tone colours of the Japanese wind instrument. Breath, either alone or mixed with the tone, is audible from *pp* to *mp*, but beyond that the tone will mask its sound unless one blows on the edge of the mouthpiece without touching it, as in Toshio Hosokawa's *Elemental Study 2*.

The first system consists of two staves and two diagrams. The top staff has notes with accidentals: $\sharp e$, b , and $\sharp b$. The bottom staff has notes with accidentals: ϕ , ϕ , and ϕ . The diagrams show fingerings and articulations. The left diagram shows a sequence of notes with fingerings C_3 , T_c , T_a , C_5 , and (Ti) . The right diagram shows a sequence of notes with fingerings C_2 , C_4 , and C_1 .

The second system shows a sequence of notes and accidentals: $\sharp e$, b , $\sharp b$, ϕ , b . The notes are: T_a , T_c , C , T_a , T_c , C_1 , T_c , C_1 , T_c , C_1 .

8va.....

The third system shows a sequence of notes and accidentals: $\sharp e$, b , $\sharp b$, ϕ , b , $\sharp b$, ϕ , $\sharp e$. The notes are: C_3 , C_4 , C_3 , C_4 , E_b , E_b , C , C_3 , C_5 .

Fig. 10.2 Altissimo register fingering chart, alto saxophone (Claude Delangle)

Overblowing

By changing the shape of the oral cavity, altering the lower-jaw pressure and forming a variety of vowel shapes, different colours and harmonics can be produced. This technique also allows the separation of the upper partials, and it is the basis of Sigurd Rascher's work on the altissimo register.

The altissimo register

Although this register will always be difficult (in terms of intonation, extreme dynamics, legato and running passages), it is now standard practice.²

Today, even without including the works dedicated to Rascher or the Rascher Quartet, there are many pieces which use this register. It permits microtones (mastering intonation is difficult), glissandi, bends and microtonal trills (and often provides some of the notes of multiphonics), although conversely a large number of other contemporary techniques are not possible here (flutter-tonguing, breath noise, circular breathing etc.). Apart from works for soloists, composers are advised to use the altissimo in moderation. Harmonic tones should not be confused with altissimo. A composer will indicate 'o' to signify the use of a colour (similar to a violin's harmonic) in the high register or during extremely quiet passages. *Xas*, the quartet by Iannis Xenakis, explores the transformation from order to chaos, using a dual texture of sweeping lines which fragment into an ever-expanding range, up to the limit of the altissimo register.

Micro-intervals, glissandi and portamenti

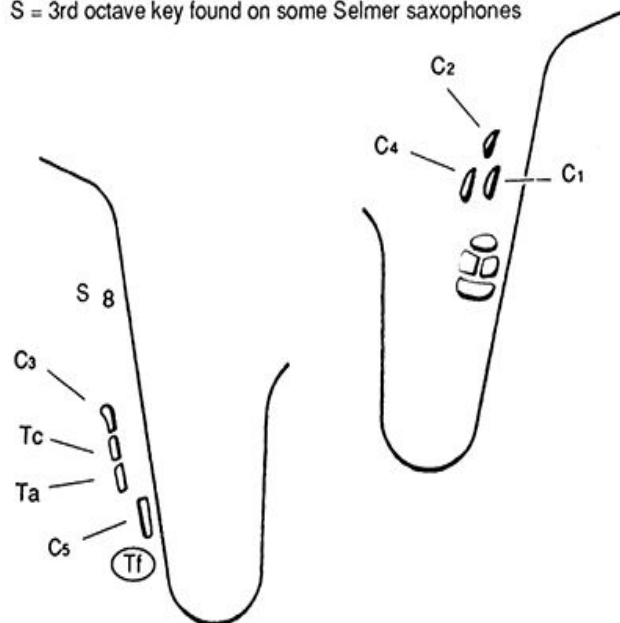
At a moderate tempo all the quarter-tones from c[#]1 are playable, although the intonation of the quarter-tone above G (written) is approximate.

Many quarter-tone trills are fairly easy and, in this respect, the tables in the book *Hello! Mr Sax* (Leduc) by Jean-Marie Londeix are very useful. Intervals less than a quarter-tone (third-tone, fifth-tone) are not all possible. Michaël Jarrell uses them very skilfully in *Résurgences* for saxophone and small ensemble. The sextet for violin, viola, cello, saxophone, horn and

clarinet by Ezra Sims, a microtonal work based on a 72-note chromatic scale (!), includes quarter-tones, sixth-tones and twelfth-tones. In *Périple*s (Paul Méfano) and *Le Frêne Egaré* (François Rossé) the microintervallic writing creates a detempered scale where the composers do not require absolute intonation. In *Four*⁵, John Cage does not indicate any micro-intervals as such but specifies that ‘intonation must be specific to each musician: a unison is therefore to be understood as a unison of differences’.

With the exception of the low register, glissando is fairly straightforward for small intervals (seconds and thirds) although beyond that portamento, as used by singers, tends to be adopted, as, for example, in Betsy Jolas’ *Fourth Episode* and *Points d’Or*. A glissando can be produced either with the lips and oral cavity or by gradual movement of the keys. It can be wide in the high register (up to the sixth harmonic) but is restricted in the lower register where it is less than a semitone. It is difficult to change registers when producing a glissando by key movement.

S = 3rd octave key found on some Selmer saxophones



Three staves of music showing fingerings for various notes. The first staff shows B $\flat$, (C $\sharp$), and B. The second staff shows (B), (C), and Tf. The third staff shows add $\frac{1}{2}$ G $\sharp$, C, and Ta.

A single staff of music showing fingerings for various notes. The notes are Ta, Tc, Tc B, Tc (Tf), C₁, B $\flat$, C₃, (B), C₃, C₅, C, B or B $\flat$, and B.

A single staff of music showing fingerings for various notes. The notes are B $\flat$, Tf, add $\frac{1}{2}$ G $\sharp$, (B), Ta, Ta, Tc, and B.

A single staff of music showing fingerings for various notes. The notes are Tc, B, C₁, C₂, C₁, C₃, C₃, C₂ (C₁), C₁, C₃, C₅, and Ta C₅.

Fig. 10.3 Quarter-tone fingering chart (Claude Delangle)

Circular breathing

An old folk-music technique common to many countries, this consists of breathing in through the nose while continuing to play with the air collected in the cheeks. With a little practice, a player can do this discreetly so that it does not detract from the music. Certain pieces make this technique essential: Betsy Jolas' *Fourth Episode*, Christian Lauba's *Etudes* and Drake Mabry's *Ceremony I*.

Articulation

Today saxophonists have learnt to master different types of articulation in the manner of the consonants to be found in human language. Composers sometimes demand silent production (clarinet attack) or conversely extremely percussive attacks, including slap-tonguing. The latter is a very dry attack produced by pressing the reed against the mouthpiece, thus creating a vacuum before the tongue is quickly withdrawn. The reed comes away from the mouthpiece violently and hits it hard, as in Alain Louvier's *Le Jeu des sept musiques*. The sound is prolonged by continuous breath in Denisov's *Sonata*. The speed of slap-tongue repetition is not very high and can be compared with 'Bartók pizzicato' on stringed instruments. For open slap-tonguing the mouth is opened thus changing the pitch (a tube open at both ends), as in *Du Sonore* by José Louis Campana. Between these two extremes, all types of attack are possible: less precise, more precise, dry, coloured by the breath, tongue noise, or even an imitation of double-reed instruments, strings or percussion.

Double- and triple-tonguing

For very rapid tonguing, the single 'tu' is not sufficient. Saxophonists use double-tonguing ('tu-ku-tu-ku', etc.) for binary rhythms and triple-tonguing ('tu-ku-tu tu-ku-tu' or 'tu-tu-ku tu-tu-ku') for ternary rhythms. In *Arabesque 3*, Ichiro Nodaïra uses this technique for timbral variation.

Playing and singing

Well known in jazz, the ‘growl’ consists of singing in a random manner when playing fairly loudly. This technique is employed in several works, although it can be improved by pitching the voice exactly, so producing precise intervals with the saxophone sound. The latter technique works very well at dynamics below *mp* – as in Taïra’s *Pénombres 6* (alto and piano) and Hurel’s *Opcit* (tenor) – and is particularly interesting when combined with subtone. In Hosokawa’s *Elemental Study 2* the instrumentalist has to sing almost permanently, creating a luminosity of sound not unlike flutter-tonguing.

Multiphonics

Using certain fingerings, several sounds can be produced simultaneously. Some of these are consonant sounds, although the majority are dissonant with the intervals of a seventh and ninth causing the clashes. *Les Sons multiples aux saxophones* (published by Salabert) by Daniel Kientzy and *Multiphonics* by Ken Dorn contain comprehensive fingering charts, although it is important for composers to collaborate directly with the instrumentalist.

Various techniques

Trumpet sounds: the mouthpiece is removed and the lips, which are placed directly on the crook, vibrate like those of a trumpeter (Christian Lauba, *Les Sept Iles*).

Playing two saxophones simultaneously: this impressive technique was used in the 1980s by Daniel Kientzy. Only the left-hand keys are employed, ideally only on the higher saxophones (Jolas’ *Points d’Or*, Savouret’s *A flanc de Bozat*).

Ram-tonguing: this very abrupt sound, also employed by flautists, is produced by tonguing strongly into the crook itself (Franco Donatoni’s *Hot*, Jean Claude Risset’s *Voilements*).

Key noise: a percussive effect obtained by slapping down the pads without blowing, producing a sort of articulation. By slapping the notes (with one or more fingers) from written $b\flat^2$ to c^2 , the instrument can be resonated at *p* or

pp, and if the mouthpiece is in the mouth the normal playing pitch can be obtained. Sometimes hitting the keys can give the impression of filtering the sound. The tone is so diminished that only the key noise seems to be left. The ‘skeleton’ of the sound remains but the texture is lost (Christian Lauba’s *Etudes*, Nodaïra’s *Arabesque 3*).

The many ways of finishing a note must also be mastered – echoes, whispered sounds, dampened or brutally interrupted notes. Reversed sound (imitating a recorded sound played backwards) is a startling example; played without attack, the sound grows until suddenly stopped by the tongue. The works of Daniel Kientzy (*L’art du saxophone* – Nova Musica CD and booklet) and Jean-Marie Londeix (*Hello! Mr Sax* – Leduc) attempt a comprehensive list of types of attack.

Contemporary saxophone teaching

Despite an unprecedented development of repertoire and technical diversification, teaching of contemporary styles and techniques has not progressed accordingly. There are, however, some works which adopt a progressive approach to techniques. The *Quinze Mosaïques* by Hubert Prati put an emphasis on changes of timbre, micro-intervals and multiphonics, and are clearly notated. In *Aphorismes* (pieces of seventy to eighty seconds), the saxophonist composer Etienne Rolin introduces young players to a variety of styles: jazz, non-European music and electro-acoustic-related techniques. He shows how multiphonics can be adapted for different uses: broken tone, tonal transformations, moving the sound from one register to another, harmonic swoops, polyphonic effects. The *Douze Etudes Complètes pour les Jeunes Saxophonistes du XXIème Siècle* by Nicolas Prost display an interesting approach to time signatures that are unusual at a junior level (5/8, 6/8, 12/8), proportional notation, different types of attack, and a thoughtful guide to multiphonics and microtonal trills.

In the United States, Ronald L. Caravan’s *Preliminary Exercises and Etudes in Contemporary Techniques for Saxophone (Introductory Material for the Study of Multiphonics, Quarter Tones and Timbre Variation)* (Dorn Publications) is a major educational work. The exercises are straightforward but give a substantial introduction to the techniques. The same author has written a book of relatively simple concert studies entitled *Paradigms I*, which help young saxophonists to become familiar with different techniques

including proportional notation and graphic improvisation. These ten short pieces contain references to modal music and blues.

The Bordeaux composer Christian Lauba has produced a major work for Leduc with his *Neuf Etudes*, divided into four books (soprano, alto, tenor and baritone). Each étude builds up to extreme virtuosity in an instrumental parameter: continuous breathing, resonance-free tone, multiphonics, slap-tonguing etc. In an essentially idiomatic language, the work gives great importance to dynamics, timbres, attacks and rhythms. According to the composer ‘the noises of keys and breath, simultaneous sounds and subtones reveal a different aspect of the instrument, or rather its true character’.³

Mention must also be made of the *Douze Etudes Modernes* by Pierre Max Dubois, *Les Douze Esquisses* of Guy Lacour, *Connexions I* by Laureau, Lejet, Rolin and Rossé (published by Billaudot); the *Sept Jeux Musicaux* of Jean-Pierre Baraglioli and Yves Queyroux introduce improvisational concepts.

Musical score for a single melodic line on a treble clef staff. The score consists of six systems of music. The first system has a long phrase with a trill on a B-flat. The second system features sixteenth-note runs with fingerings 6, 6, 6, 6 and dynamics *pp* and *ppp*. The third system continues with similar runs and fingerings 6, 6, 7, 6, ending with a trill on a B-flat. The fourth system includes a "bisb." (biscia) trill, a "subtone" (subito) trill, a "vib. 4" (vibrato 4) trill, a "sub." (subito) trill, and a "4'" (four triplets) trill. The fifth system contains several trills with durations like 4'', 2,5'', x'', 2'', and x''. The sixth system ends with a trill on a B-flat, a "bisb." trill, and a long sustained note marked 10''. Dynamics include *p*, *pp*, *ppp*, and *pppp*. Performance instructions include "assez libre" and "very free".

Fig. 10.4 Extract from *Bat* for baritone saxophone (1996) by Christian Lauba

As educational works *per se* are thin on the ground, an educationally structured repertoire is vital.

Rhythmic freedom

Alain Louvier, *Hydre à Cinq Têtes* (second year); Graciane Finzi, *De l'un à l'Autre* (third year); Edith Lejet, *Trois Préludes* (fourth year); Christophe Havel, *Trois Rides sur l'Horizon* (fifth year); Alain Louvier, *Cinq Ephémères* (sixth year).

Microtonal trills and micro-intervals

Edith Lejet, *Trois Préludes* (fourth year); Thérèse Brenet, *Phoenix* (fifth year); Alain Louvier, *Cinq Ephémères* (sixth year).

Multiphonics

Christophe Havel, *Trois Gestes*; Thérèse Brenet, *Phoenix*; Ryo Noda, *Requiem*; Patrick Choquet, *Aires*.

Flutter-tonguing

Marie-Hélène Fournier, *Quatre Duos*; Edith Lejet, *Jade for Alto Saxophone and Percussion*.

Subtone

Marie-Hélène Fournier, *Hemlock*.

Glissandi and portamenti

Marie-Hélène Fournier, *Sétiocétime*; Ryo Noda, *Improvisations*.

Slap-tonguing

José-Luis Campana, *Pezzo Per Claudio*; François Rossé, *Lobuk Constrictor*.

Double-tonguing, growling, singing/playing, harmonics

Javier De La Vega, *Mosaicos para saxofono contemporaneo*; Serge Bertocchi, *Techniques du Saxophone*.

In his book *Les Etudes contemporaines pour le saxophone et les nouvelles techniques*, Vincent David assesses the study books and educational repertoire that make use of contemporary techniques, as well as analysing in detail the works of Lauba, Grisey, Stockhausen and Berio. There is also a series of eight studies giving an introduction to microtonality by Bruno Mantovani.

Midi wind instruments

RICHARD INGHAM

The treatment of sound externally to the saxophone follows a natural course, beginning with the pure acoustic developments of the instrument – conical tube design, mouthpiece chamber matching and reed choice. Before the advent of amplification, the musician would be concerned with note and tone production as the instrument produced them. When sound amplification is available for those who choose to use it, a new realm of possibilities emerges (although it should be remembered that room acoustics have always been taken into account by musicians as an external influencing factor on the sound as it is perceived by the listener).

From the basic reverberation effect (as used by Charlie Parker with Slim Gaillard in the 1940s, and Earl Bostic in the 1950s), further developments were minimal until the acceptance of electronic sound creation and transformation. In the 1970s, when sound synthesis became generally available, suggesting new timbral concepts, more elaborate treatment devices began to be used in rock, jazz-rock and contemporary classical music – for example delay, ring modulation and pitch shifting (octivider). It was at this point that saxophonists began to experiment with sound transformation, and players such as Andy Mackay, Michael Brecker, Jan Garbarek and Sal Gallina were looking for new textures and timbres.

Eventually, much of this experimentation culminated in the invention of the Lyricon, an electronic instrument which retains the essential fingering system of the saxophone, yet acts as a controller serving a synthesiser unit. The Lyricon was devised and first produced by Bill Bernardi and Roger Noble in the USA in 1974, and operates using a transducer system to convert wind pressure into synthesizer control information. It was used by Tom Scott, Wayne Shorter, Kenny Gee, John Walters and David Roach amongst others. At the same time, Nyle Steiner was developing his Electronic Valve Instrument (EVI), which was later followed by the Electronic Wind Instrument (EWI).

After the initial commercial success of the Yamaha DX7 keyboard synthesizer from 1983, a breath controller was produced as a means of gaining more expressive quality from the synthesizer; the keyboard player holds a note or chord, and by means of breath pressure alters the volume of the sound. Whilst quite rudimentary, the potential of this was not lost on the Yamaha company, who produced their WX7 midi wind controller in 1987 (MIDI = Musical Instrument Digital Interface). At almost exactly the same time, the Akai company, who had taken over Nyle Steiner's operations,

released the EWI/EVI as commercial products. Both these instruments, although very different, can be played by saxophonists, and both make use of breath control and woodwind articulation for expressive performance of synthesized sounds. The introduction of midi enabled the performer to use a range of electronic equipment, rather than being limited to, and governed by, one sound source.

The WX7 is a controlling device, which will work in conjunction with any midi receiving module. The instrument transmits on two midi channels, and offers pitch choice through a range of over six octaves based on slightly adapted saxophone/recorder fingering. This is naturally the woodwind equivalent of depressing the keys on a keyboard. The wind and lip sensors, however, convert (via transducers) breath and lip alterations into midi messages, enabling the performer to use typical wind performance techniques with similar responses to acoustic instruments (for example an increase of air gives higher volume; adjustment of lip pressure produces pitch alteration for bends and vibrato). An incidental use of these facilities is that wind and lip control can easily be assigned by midi to operate any parameter – thus air increase may produce pitch change instead of volume change. Control by the tongue is crucial, not only in differentiating between a successful and a tasteless performance, but also in grading the resultant tone colour very finely. The tongue attack (equivalent of ‘keyboard velocity’) can produce a very subtle opening to a note (or can ruin it) and, like an acoustic instrument, can produce anything from a very light to a very heavy attack, the reasons for which will be dependent on style and context. The stock-in-trade techniques taken for granted by wind players as part of their natural training now become of enormous importance in emphasising the expressive function within electronic performance.

Other features of the WX7 midi wind controller and later models WX11 and WX5 include program change keys, a pitch wheel and a drone facility (whereby one note may be held while a separate line is played simultaneously). The mouthpiece is designed to be similar to saxophone and clarinet mouthpieces, although no sophisticated embouchure is necessary (due to the ‘reed’ not vibrating acoustically) and options are available to the player when setting the air resistance (whether the player wishes to feel the air passing through the instrument when blowing, as with recorder and flute, or feel resistance in the mouthpiece, as with the clarinet). The wind and lip

sensors are set by the individual player to correspond to their particular breath and lip strength and sensitivity.

The EWI is a completely different looking beast to the WX7, but produces a broadly similar result, inasmuch as the two can be interchangeable as controllers. The EWI has been described as a ‘chairleg’ design, but the almost deliberate non-chic appearance belies its sophistication. Whereas the WX7 is designed to feel like a saxophone as far as the mouthpiece and fingers are concerned, the EWI has a flattened teat for a mouthpiece, and touch sensors for a fingering system. Vibrato is achieved by squeezing the mouthpiece, which is relatively straightforward for a wind player, but finger control requires delicate co-ordination. The system works very well, once the player has learnt not to touch the sensors unless required – an accidental slight finger movement will trigger a different note (unlike the WX7 which requires key depression). Left-thumb rollers at the back of the instrument operate octave changes, and a glide plate for the right thumb allows portamento. Whereas the WX7 controls a midi tone generator directly, the EWI requires its own sound module, incorporating features such as transposition and chord production. This module can be used in conjunction with any midi receiving device, thus the available sounds are largely the same on both instruments – limited only by the tone generator and creativity of the programmer.

When the instruments were launched in 1987, each company had an outstanding exponent of the instrument to ease over the uncertainty of new technology. Yamaha presented a series of remarkable demonstrations given by Sal Gallina, whose concept of orchestral sound from a synthesizer was matched by his inventive flair. His programmed sounds were largely imitative of orchestral, jazz and rock instruments and sections. By clever use of the drone feature, he was able to create many layers of sound and texture. His finale was a heavy metal guitar feature. It should be noted here that Gallina’s impact was possible due to the expressive nature of the wind controller. The enormous range of sensitivity and tonal control available to the wind player could now be used to control synthesized sound in a way not heard before. This was a notable step forward, not just for wind instrument technology, but also for synthesized sound, its use and perception.

Michael Brecker’s *Magnetic* album with Steps Ahead (1986) introduced the ‘Steiner EWI’ on several tracks. This is the instrument which Akai manufactured the following year, and Brecker’s version of Ellington’s *In a*

Sentimental Mood on this album became a classic. The sounds he employed were not extensive in number (they were largely ‘created’ – non-imitative), but his artistic use of them was as breathtaking as his saxophone playing. He was able to exploit the eight-octave range and new fingering system to point to dramatic new possibilities for the wind player. By controlling the Oberheim Xpander synthesizer via the EWI he could operate filter changes through breath control, giving a very exciting yet subtle shading to his playing. On *Michael Brecker* (his first solo album), released in 1987, he used chordal playing to great effect in *Original Rays*. The following year *Itsbynne Reel* on the album *Don't Try This at Home* was an excellent example of his use of layered voicings, in this case a string/flute sound. Through the work of Gallina and Brecker, saxophonists were able to hear the full range of potential at the highest standard of performance – both the range of imitative sound possibilities with Gallina, and the solo excursions of Brecker.

Initial response to the instruments was varied – some players embraced the new technology and incorporated it into their work, others rushed to invest in the latest thing and wondered what they might do with it; some found themselves unwilling to cross the technology barrier, and others did not see it as any relation to the acoustic instrument at all. One of the initial marketing problems was that the instruments were sold in stores designed to sell keyboards, because of the equipment required, yet those most likely to buy them were woodwind players who needed the long term-expertise and back-up of specialist woodwind suppliers. The instruments did sell in their numbers, however, and despite its youth and the requirement for rethinking, the midi wind controller has already been involved in and been responsible for much genuine artistic creativity.

Concert works were written as early as 1988, including the *Concerto for Electronic Wind Instrument and String Orchestra*, premièred by Frederick Hemke at the 9th World Saxophone Congress in Tokyo/Yokohama. John Samphen was the soloist for *In Two Worlds* by Morton Subotnick, where the soloist moves between wind controller and saxophone; as well as driving sounds, live electronics are controlled by the wind controller via computer during the performance. The 10th World Saxophone Congress in Pesaro, Italy (1992), contained two WX7 recitals, by Bruce Ronkin and Richard Ingham, featuring *Future Echoes from the Ancient* by William Moylan, *Whale* by Terry Davies, *Junctions* by Allan Greenwood and *Still Life* by Richard Ingham. Debra Richtmeyer gave performances of *Proper Behavior*

by Steven Everett in the USA and Europe in 1994; this work requires the soloist to play WX7, saxophone and to speak – all the material is combined to produce a very fine example of interactive electronics.

The early concert items, the album session work by Sal Gallina and the solo output by Michael Brecker all demonstrate the breadth of use of this electronic instrument. David Roach, using the Akai EWI and latterly the Yamaha WX11, has inspired several composers at the Royal National Theatre in London to write for the instrument, notably Dominic Muldowney (*The Pied Piper* and *Under Milkwood*) and Paddy Cunneen (*Millennium* and *Perestroika*). Roach was one of the few in the UK to own a Lyricon, and his marvellous programming skills followed him on to the EWI–Oberheim Xpander combination, later adding the WX11 controller–VL1m virtual acoustic tone generator to his list. Each sound he created, along with customised effects, was a demonstration of possibilities, as well as being integral to the drama. Perhaps it is in the theatre that this frightening instrument can best display its potential, with a full range of imitative and created voices. Other theatrical midi events in Britain have included Terry Davies's music for *Whale* at Sheffield Crucible, and his *Kes* at Bolton Octagon.



Fig. 10.5 Debra Richtmeyer with Steven Everett (composer of *Proper Behavior*), Third British Saxophone Congress, 1994

In summing up the potential and the problems, we return to one of the inventors of the Lyricon, Bill Bernardi:

A synthesizer's ability to replace or surpass conventional musical instruments as art forms will rely heavily on how well they are able to translate, in natural fashion, the player's expression into sound . . . an acoustic instrument is basically a highly sensitive and articulate control system coupled to a human player who is capable of transmitting highly subtle degrees of control into the acoustic sound generating system. The awesome complexity lies with the ability of the instrument to interface and relate to the player.¹